

KANCHARLA TRIDEV SANJAY

AI Engineer — Machine Learning — Generative AI — Backend Systems

📍 Chennai, Tamil Nadu, India 📞 +91 9949128732
✉️ tridevsanjay06@gmail.com [in LinkedIn](#) [GitHub](#)

PROFESSIONAL SUMMARY

Final-year B.Tech student in Artificial Intelligence & Machine Learning with hands-on experience building machine learning, Generative AI and full-stack software systems. Skilled in Python, TensorFlow, Scikit-learn, FastAPI, AWS and backend engineering. Experienced in developing ML pipelines, model evaluation workflows, prediction systems, REST APIs and AI-powered applications. Passionate about building scalable AI systems, LLM applications and production-ready machine learning infrastructure.

MACHINE LEARNING ENGINEERING

- Built end-to-end machine learning workflows including data preprocessing, feature engineering, model training, evaluation and deployment.
- Developed prediction and classification models using TensorFlow and Scikit-learn for real-world applications.
- Designed REST APIs and inference services for integrating ML models into production-ready applications.
- Worked with large datasets using Pandas, NumPy and data transformation pipelines.
- Implemented model evaluation strategies, experimentation workflows and iterative optimization techniques.
- Built Generative AI applications using LLMs, prompt engineering and retrieval-based architectures.

SELECTED PROJECTS

VideoForge AI [\[GitHub\]](#)

Python • FastAPI • LLMs • RAG • Next.js • AWS

- Built an AI-powered platform that converts product documentation and release notes into scripts, storyboards and marketing assets.
- Integrated NVIDIA NIM and LLM APIs to generate contextual content using prompt engineering workflows.
- Developed FastAPI inference services and backend APIs for scalable AI processing.
- Implemented retrieval-based pipelines and structured context generation to improve output quality.
- Designed modular AI workflows supporting experimentation, evaluation and rapid iteration.

Cloud EV Power Sharing System [\[GitHub\]](#)

Python • Node.js • MongoDB • AWS • Socket.IO

- Developed a cloud-based EV energy sharing platform enabling dynamic energy allocation and transaction management.
- Designed scalable backend services for user management, energy pooling and real-time updates.
- Built analytics modules and data processing workflows for utilization forecasting and decision support.
- Implemented location-aware charging station discovery and cloud deployment architecture.
- Optimized performance through modular service design and efficient database operations.

AI-Powered Carbon Footprint Calculator [\[GitHub\]](#)

Python • Scikit-learn • Pandas • React • REST APIs

- Built a machine learning prediction system achieving 85% accuracy for carbon footprint estimation.
- Performed data preprocessing, feature engineering and model evaluation using Scikit-learn.
- Developed predictive analytics APIs enabling real-time environmental impact estimation.
- Integrated ML models with a full-stack application delivering sustainability insights.
- Managed complete ML lifecycle including data preparation, training, testing and deployment.

EXPERIENCE

Backend Developer Intern – Hewlett Packard Enterprise (HPE)

2025

- Developed RESTful APIs using Java and Spring Boot for enterprise backend systems.
- Designed modular services supporting scalable application architectures.
- Worked with relational databases, service integrations and backend optimization techniques.
- Applied software engineering best practices including debugging, testing and API validation.
- Collaborated using Git, Postman and Agile development methodologies.

EDUCATION

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

Expected 2026

B.Tech in Artificial Intelligence & Machine Learning

CGPA: 8.0/10

TECHNICAL SKILLS

Programming: Python — Java — SQL — JavaScript — TypeScript

Machine Learning: Scikit-learn — TensorFlow — Classification — Regression — Model Evaluation — Feature Engineering — Predictive Analytics

Generative AI: LLMs — RAG — Prompt Engineering — Embeddings — Vector Search — AI Agents

Data Engineering: Pandas — NumPy — Data Pipelines — Data Preprocessing — Feature Pipelines

Backend Engineering: FastAPI — Node.js — Spring Boot — REST APIs — Microservices

Cloud & Infrastructure: AWS — Model Deployment — API Deployment — Git — GitHub

Databases: MongoDB — MySQL

Core Computer Science: Data Structures & Algorithms — OOP — DBMS — Operating Systems — System Design

CERTIFICATIONS

- AWS Solutions Architecture Virtual Experience Program – Forage
- Artificial Intelligence Professional Certification – OneRoadmap
- Generative AI Certification – Infosys Springboard
- Power BI for Data Analytics Certification
- Microsoft Excel for Business Analytics Certification

ADDITIONAL INFORMATION

- Built and deployed multiple machine learning and AI-powered applications from development to deployment.
- Experience developing backend systems and inference APIs supporting AI workloads.
- Strong understanding of ML model evaluation, experimentation and iterative optimization.
- Active GitHub portfolio showcasing AI engineering, machine learning and software development projects.
- Passionate about scalable AI systems, developer tools, LLM applications and production machine learning.